

9. Best Fit Contiguous Memory Allocation

FACE Prep

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Best Fit Contiguous Memory Allocation

Completed

Write a C program to implement the Best Fit Contiguous Memory Allocation technique. The program should accept the sizes of memory blocks and the sizes of processes as input.

For each process, the program searches all available memory blocks and allocates the smallest block that is sufficient to satisfy the process size. If no suitable block is available, the process is marked as not allocated. The program should display the allocation details for each process and calculate the internal fragmentation.

Input Format

1. The first line contains an integer representing the number of memory blocks.
2. The second line contains the sizes of the memory blocks.
3. The third line contains an integer representing the number of processes.
4. The fourth line contains the sizes of the processes.

Output Format

• For each process, display whether it is allocated or not.
• If allocated, display the block number, block size, and internal fragmentation.
• If not allocated, display that the process could not be allocated.

Example 1
Sample Input 1
5
100 500 200 300 600
4
212 417 112 426

Select Language: C (GCC 9.2.0)

```
1 #include <stdio.h>
2
3 int main() {
4     int m, n;
5
6     printf("Enter number of memory blocks:\n");
7     scanf("%d", &m);
8
9     int blockSize[m];
10    int remaining[m];
11
12    printf("Enter size of each block:\n");
13    for (int i = 0; i < m; i++) {
14        scanf("%d", &blockSize[i]);
15        remaining[i] = blockSize[i];
16    }
17
18    printf("Enter number of processes:\n");
19    scanf("%d", &n);
20
21    int processSize[n];
22    int allocation[n];
23    int fragment[n];
```

Test Cases

Run Sample Cases

Run All Cases

Custom Test Case

Custom Test

Sample Test Case

Test Case - 1

Test Case - 2

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Select Language: C (GCC 9.2.0)

```
22 int allocation[n];
23 int fragment[n];
24
25 for (int i = 0; i < n; i++) {
26     allocation[i] = -1;
27     fragment[i] = 0;
28 }
29
30 printf("Enter size of each process:\n");
31 for (int i = 0; i < n; i++) {
32     scanf("%d", &processSize[i]);
33 }
34
35 for (int i = 0; i < n; i++) {
36     int bestIndex = -1;
37
38     for (int j = 0; j < m; j++) {
39         if (remaining[j] >= processSize[i] && (bestIndex == -1 || remaining[j] < remaining[bestIndex])) {
40             bestIndex = j;
41         }
42     }
43
44     if (bestIndex != -1) {
45         allocation[i] = bestIndex;
46         remaining[bestIndex] -= processSize[i];
47         fragment[i] = remaining[bestIndex];
48     }
49 }
```

Test Cases

Run Sample Cases

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Example 1

Sample Input 1

```
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100 500 200 300 600
4
212 417 112 426
```

Select Language: C (GCC 9.2.0)

```
48     fragment[i] = remaining[bestIndex] - processSize[i];
49     remaining[bestIndex] -= processSize[i];
50 }
51 }
52
53 for (int i = 0; i < n; i++) {
54     if (allocation[i] != -1) {
55         printf("Process %d of size %d is allocated to Block %d of size %d with Fragment %d\n",
56             i + 1,
57             processSize[i],
58             allocation[i] + 1,
59             blockSize[allocation[i]],
60             fragment[i]);
61     } else {
62         printf("Process %d of size %d is not allocated\n",
63             i + 1,
64             processSize[i]);
65     }
66 }
67
68 return 0;
69 }
70 }
```

Test Cases

Run Sample Cases

Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1

> Test Case - 2